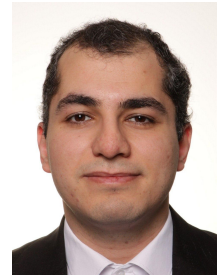


Amirhossein Bayat

Ph.D. Candidate in Mechanical Engineering
University of Southern Denmark

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Profile

Ph.D. candidate in Mechanical Engineering at the University of Southern Denmark, working on topology optimisation for high-heat-flux cooling and turbulent fluid-flow problems. My research combines gradient-based optimisation, finite-element modelling, adjoint sensitivity analysis, conjugate heat transfer, and turbulent-flow simulation to develop efficient computational design methods for advanced thermal-fluid systems.

Education

University of Southern Denmark

Ph.D. in Mechanical Engineering

Odense, Denmark

2024–Present

- Supervisor: Joe Alexandersen.
- Thesis topic: Topology optimisation for high-heat-flux cooling.

Sharif University of Technology

M.Sc. in Mechanical Engineering

Tehran, Iran

2021–2023

- Supervisor: Ali Hakkaki Fard.
- Thesis title: Solar PV/T collectors enhanced with PEM electrolytic cells.
- CGPA: 18.25/20 (4/4).

Shahid Beheshti University

B.Sc. in Mechanical Engineering

Tehran, Iran

2015–2019

- CGPA: 17.62/20 (3.8/4).

Selected Research Experience

Topology optimisation for high-heat-flux cooling

University of Southern Denmark

Ph.D. research project

2024–Present

- Developing computational methods for thermal-fluid topology optimisation with emphasis on cooling performance and manufacturable engineering designs.
- Working with finite-element formulations, adjoint-based sensitivities, PDE filtering, and MMA-based optimisation workflows.
- Extending optimisation methods towards turbulent and conjugate-heat-transfer problems relevant to advanced cooling applications.

Solar PV/T collectors enhanced with PEM electrolytic cells

Sharif University of Technology

M.Sc. thesis project

2021–2023

- Modelled coupled thermal-energy systems and investigated strategies for improving PV/T collector performance.
- Combined numerical modelling, thermodynamic analysis, and engineering design considerations for renewable-energy applications.

Selected Projects

ASME K-16 Heat Transfer in Electronic Equipment: 2025–2026 Cold Plate Competition

University of Southern Denmark

Heat-sink / cold-plate design project

2025–2026

- Designed and analysed a compact heat-sink/cold-plate concept for high-heat-flux electronic cooling.
- Combined thermal-fluid simulation and design optimisation ideas to improve heat removal and flow distribution.

Transient temperature modelling of a photovoltaic-thermal cell

Sharif University of Technology

Tehran, Iran

Feb. 2023–Mar. 2023

- Investigated the outlet-water temperature of a PV/T collector during operating hours.
- Studied the use of phase-change materials and nanofluids to improve simulated cell performance.
- Wrote UDF codes to define time-dependent solar radiation, wind speed, and ambient temperature.

Heat-flux estimation using inverse heat-transfer methods

Sharif University of Technology

Tehran, Iran

Feb. 2023–Mar. 2023

- Developed Fortran code for inverse heat-transfer calculations and noise-error analysis.
- Created an ANSYS–MATLAB coupled simulation workflow to solve the inverse problem.

Publications and Manuscripts

Density-based topology optimization for Navier–Stokes flow with free-slip boundary conditions

Peer-reviewed journal article

Structural and Multidisciplinary Optimization, 69(1), Article 23

2026

- Amirhossein Bayat, Hao Li, and Joe Alexandersen.
- DOI: 10.1007/s00158-025-04212-7.

Density-based topology optimization for turbulent fluid flow using the standard $k-\epsilon$ RANS model with wall-functions imposed through an implicit wall penalty formulation

Preprint / manuscript

arXiv:2601.02202

2026

- Amirhossein Bayat, Hao Li, and Joe Alexandersen.
- Focus: turbulent fluid-flow topology optimisation using RANS modelling and implicit wall-function treatment.

Research Interests

Core areas

Fluid mechanics, turbulent fluid simulation, heat transfer, topology optimisation, conjugate heat transfer, and high-heat-flux cooling.

Methods

Finite element method, adjoint sensitivity analysis, PDE-constrained optimisation, RANS turbulence modelling, wall-function modelling, and MMA-based optimisation.

Emerging interests

Microfluidics, particle separation, acoustofluidics, and thermal-fluid design for fusion-energy applications.

Teaching Experience

Structural Engineering, M.Sc. course

Teaching Assistant

University of Southern Denmark

Spring 2025

- Supported master's-level teaching activities, exercises, and student learning in structural engineering.

Advanced Finite Element Method, M.Sc. course

Teaching Assistant

University of Southern Denmark

Fall 2025

- Supported master's-level teaching activities, exercises, and student learning in advanced finite-element analysis.

Heat-transfer experiments for B.Sc. students

Laboratory and Experimental Teaching Support

University of Southern Denmark

2024–2026

- Responsible for preparing, supporting, and guiding undergraduate heat-transfer laboratory experiments.

Conferences and Symposia

World Congress of Structural and Multidisciplinary Optimization (WCSMO)

Oral presentation

Kobe, Japan

2025

- Presented research on density-based topology optimisation for fluid-flow problems.

WCCM–ECCOMAS 2026

Oral presentation

Munich, Germany

2026

- Presentation at the 17th World Congress on Computational Mechanics and 10th European Congress on Computational Methods in Applied Sciences and Engineering.

DCAMM Internal Symposium

Poster presentation

Denmark

2024

- Presented Ph.D. research progress in applied mathematics and mechanics.

DCAMM Internal Symposium

Oral presentation

Denmark

2026

- Presented research progress in topology optimisation and thermal-fluid modelling.

Granted Funds

THOMAS B. THRIGES FOND – WCSMO Congress, Kobe, Japan

Grant reference: 5104-2411

Travel grant, DKK 20,000

Nov. 2024–Jul. 2025

- Awarded support for participation in the Congress of Structural and Multidisciplinary Optimization.
- Funding agency: THOMAS B. THRIGES FOND.

DANfusion – WCCM–ECCOMAS 2026, Munich, Germany

DANfusion, Denmark

Travel support, DKK 12,000

2026

- Awarded funding support for participation in WCCM–ECCOMAS 2026, connected to fusion-energy research.
- DANfusion travel support is intended for students and scientists at Danish universities in relation to fusion-energy research.

Technical Skills

Programming MATLAB, Python, Fortran, FreeFEM++, Firedrake.

Simulation COMSOL Multiphysics, ANSYS Fluent, Abaqus, Tecplot, PipeNet.

Numerical methods FEM, turbulent-flow simulation, topology optimisation, adjoint sensitivities, and PDE filtering.

CAD and tools SolidWorks, CATIA, Inventor, AutoCAD, LaTeX/Overleaf, Microsoft Office.

Professional Scientific writing, documentation, teamwork, problem solving, and research presentation.

Languages

English Professional proficiency, C1.

Persian Native proficiency.

Danish Elementary proficiency, A2.

References

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